
MODULE *TrafficLight*

EXTENDS *Naturals*

VARIABLES *TrafficLight*, *turn*

$States \triangleq \{\text{"green"}, \text{"red"}\}$

$vars \triangleq \langle TrafficLight, turn \rangle$

$Init \triangleq$
 $\wedge turn = 1$
 $\wedge TrafficLight = [i \in 0 \dots 1 \mapsto \text{"red"}]$

$TypeOK \triangleq$
 $\wedge \forall i \in 0 \dots 1 : TrafficLight[i] \in States$
 $\wedge turn \in \{0, 1\}$

$OtherTL(a) \triangleq (a + 1) \% 2$

$Control \triangleq$
 $\wedge TrafficLight[turn] = \text{"green"}$
 $\wedge turn' = OtherTL(turn)$
 $\wedge \text{UNCHANGED } TrafficLight$

$SwitchToRed \triangleq$
 $\wedge TrafficLight[OtherTL(turn)] = \text{"green"}$
 $\wedge TrafficLight' = [TrafficLight \text{ EXCEPT } ![OtherTL(turn)] = \text{"red"}]$
 $\wedge \text{UNCHANGED } turn$

$SwitchToGreen \triangleq$
 $\wedge TrafficLight[OtherTL(turn)] = \text{"red"}$
 $\wedge TrafficLight' = [TrafficLight \text{ EXCEPT } ![turn] = \text{"green"}]$
 $\wedge \text{UNCHANGED } turn$

$Next \triangleq$
 $\vee Control$
 $\vee SwitchToGreen$
 $\vee SwitchToRed$

$NeverBothGreen \triangleq TrafficLight[0] = \text{"red"} \vee TrafficLight[1] = \text{"red"}$

$SwitchOnlyAfterGreen \triangleq \Box [TrafficLight[turn] = \text{"red"} \Rightarrow turn' = turn]_{vars}$

$SwitchNotDirectlyToGreen \triangleq \Box [\forall i \in 0 \dots 1 :$
 $\wedge TrafficLight[i] = \text{"green"}$
 $\wedge TrafficLight[OtherTL(i)] = \text{"red"}$
 $\Rightarrow$
 $\vee TrafficLight' = TrafficLight$
 $\vee TrafficLight' = [TrafficLight \text{ EXCEPT } ![i] = \text{"red"}]]_{vars}$

$SwitchToGreenWhoseTurnItIs \triangleq \Box [\forall i \in 0 \dots 1 :$
 $\wedge TrafficLight[i] = \text{"red"}$

$$\begin{aligned}
& \wedge \text{TrafficLight}[\text{OtherTL}(i)] = \text{"red"} \\
& \Rightarrow \\
& \vee \text{TrafficLight}' = \text{TrafficLight} \\
& \vee \text{TrafficLight}' = [\text{TrafficLight} \text{ EXCEPT } ![\text{turn}] = \text{"green"}]]_{vars} \\
\text{AlwaysEventuallyGreen} & \triangleq \square(\forall i \in 0 \dots 1 : \text{TrafficLight}[i] \triangleq \text{"red"} \Rightarrow \diamond \text{TrafficLight}[i] = \text{"green"})
\end{aligned}$$

Sei *Ampel 1* am *Zug*. Dann kann *Ampel 2* erst dann dran kommen, nach dem *Ampel 1* auf *Grün* geschaltet wurde.

\ * Modification History
\ * Last modified *Wed Mar 20 15:38:46 CET 2024* by *moritz*
\ * Created Sat *Dec 30 18:44:33 CET 2023* by *moritz*